

Year 11 Revision booklet

Biology

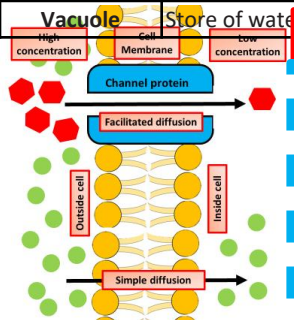
Easter holidays Revision

Instructions:

Use the knowledge organizer to help you answer the questions on the revision sheets.

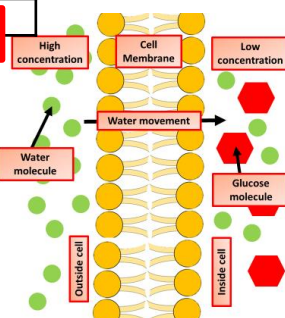


Cells & Organelles.	
Organelle	Function
Cytoplasm	Where chemical reactions take place
Nucleus	Contains genetic material (genes & chromosomes) & controls cell activity.
Cell membrane	Controls what enters & leaves the cell
Mitochondria	Organelles that contain the enzymes for respiration, and where most energy is released in respiration.
Ribosomes	Tiny structures where protein synthesis occurs.
Chloroplast	Site of photosynthesis in plants
Cell wall	Strengthen plant cells



Diffusion

Diffusion is the movement of a substance from an area of high concentration to an area of low concentration. Diffusion happens in liquids and gases because their particles move randomly from place to place. Diffusion is an important process for living things; it is how substances move in and out of cells.

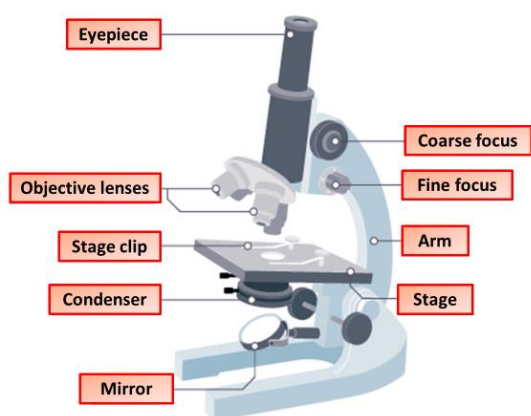


Osmosis

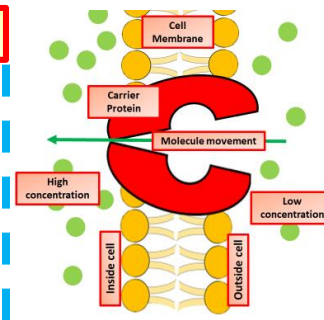
Osmosis is the movement of water or other solvent through a plasma membrane from a region of low solute concentration to a region of high solute concentration. Osmosis is passive transport, meaning it does not require energy.

Using microscopes

1. Cut open an onion
2. Use forceps to peel a thin layer of epidermis from the inside
3. Lay the layer of epidermis on a microscope slide
4. Add a drop of iodine solution to the layer
5. Carefully place a cover slip over the layer
6. Examine the slide under a microscope
7. Draw what you can see



$$\text{Magnification} = \frac{\text{Image size}}{\text{Actual size}}$$



Active transport

active transport is the movement of molecules across a membrane from a region of their lower concentration to a region of their higher concentration—against the concentration gradient. Active transport requires cellular energy to achieve this movement.

Mitosis

Stage 1: The cell spends most of its life in the interphase. During this phase the cell grows to its maximum size and performs its normal functions.

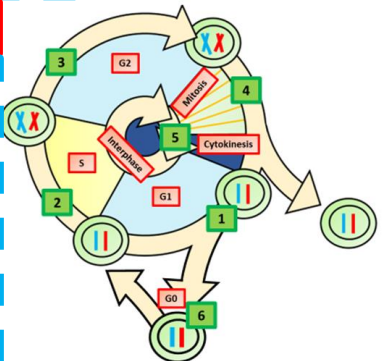
Stage 2: The DNA condenses into chromosomes (human cells have 46 chromosomes – 23 from your father and 23 from your mother). Each chromosome eventually can be seen to consist of two strands or chromatids joined at a central centromere in an X shape.

Stage 3: The nuclear membrane disappears. Spindle threads form between the poles.

Stage 4: Chromosomes lie on the equator of the cell. Each chromosome is attached to the spindle microfibers by its centre. The chromosomes appear in a straight line across the middle of the cell.

Stage 5: The centre of the chromosome splits. Each chromosome divides into two sister chromatids. Each chromatid is moved to opposite poles of the cell by the shortening of the spindle fibres. Chromatids (now called daughter chromosomes) gather at opposite poles of the cell.

Stage 6: A nuclear membrane forms around each of the daughter chromosomes that have gathered at the poles. The daughter chromosomes uncoil.



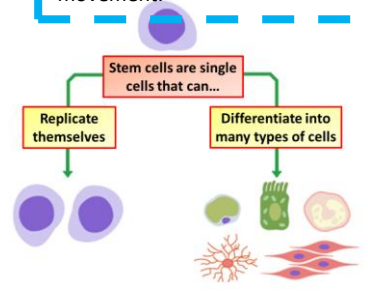
Cell cycle

In cells with a nucleus, as in eukaryotes, the cell cycle is also divided into two main stages: interphase and the mitotic (M) phase (including mitosis and cytokinesis). During interphase, the cell grows, accumulating nutrients needed for mitosis, and undergoes DNA replication preparing it for cell division.

Stem cells

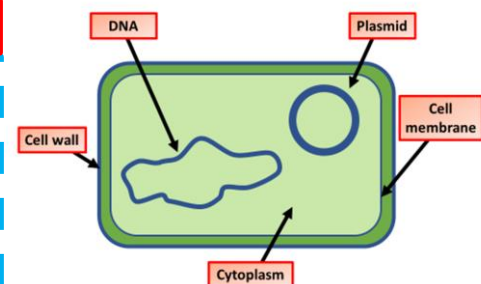
Stem cells from a recently dead embryo can be grown in special solutions.

- Stem cells from an embryo can grow into any type of tissue.
- Stem cells may grow out of control, to form cancers.
- Large numbers of stem cells can be grown in the laboratory.
- Stem cells may be used in medical research or to treat some human diseases.
- Patients treated with stem cells need to take drugs for the rest of their life to prevent rejection.
- Collecting and growing stem cells is expensive

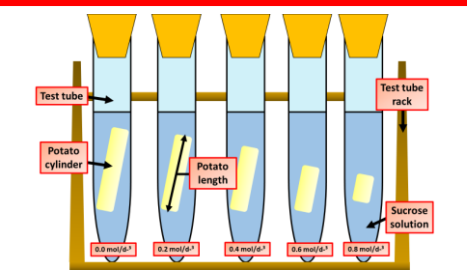


Prokaryotic & Eukaryotic Cells

Eukaryotic cells contain membrane-bound organelles, including a nucleus. Eukaryotes can be single-celled or multi-celled, such as you, me, plants, fungi, and insects. Bacteria are an example of prokaryotes. Prokaryotic cells do not contain a nucleus or any other membrane-bound organelle.

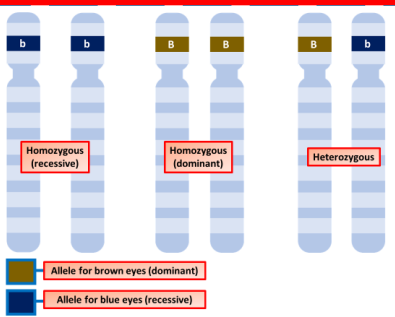


Osmosis required practical



- Prepare a range of sucrose (sugar) solutions. The concentration of a solution is measured in moles per cubic decimetre written as mol dm⁻³. For example, in this experiment your range could be from 0.2 mol dm⁻³ to 1.0 mol dm⁻³. A 1.0 mol dm⁻³ solution of sucrose will contain up to 342 g of sucrose per dm⁻³. A 1.0 mol dm⁻³ solution of a substance contains one mole of the substance per dm³ of a solution, or one mole per litre of solution.
- Set up a series of boiling tubes with each of these solutions. Also, set up one containing distilled water. This will have a concentration of sucrose of 0.0 mol dm⁻³ and will act as the control in the experiment.
- Make sure each tube is labelled with the concentration.
- Carry out the investigation. Prepare a blank results table before you begin. Make sure when weighing the potato cylinders, that their masses are not mixed up when recording them. Each cylinder will have a different mass before and after the investigation.
- For each sucrose concentration, repeat the investigation for several potato cylinders. This allows you to make the experiment more repeatable – not all potato cylinders might behave in the same way. Making a series of repeat experiments means that any anomalous results can be identified and ignored when a mean is

Chromosomes



- Chromosomes carry genetic information in a molecule called DNA.
- A type of cell division called mitosis ensures that when a cell divides each new cell produced has the same genetic information.
- DNA exists in a cell's nucleus within structures called **chromosomes**. Each section of a chromosome that contains the code for the production of a particular protein is called a **gene**.

Specialised cells in plants

	Cells of the...	Specialised to...
Leaf	Palisade mesophyll	Carry out photosynthesis
	Spongy mesophyll	Allow gases to circulate for the exchange of gases between the leaf and the environment, carry out some photosynthesis
	Guard cells	Open and close to control the exchange of gases – carbon dioxide, water vapour and oxygen
Phloem	Sieve tubes	Transport products of photosynthesis, including sugars and amino acids, from the leaf to where they are needed
	Companion cells	Provide the energy required for transporting substances in sieve tubes
Xylem	Xylem vessels	Transport water and dissolved minerals from the roots, up the plant
Growing points	Meristem	Produce new cells as they divide

Specialised cells in animals

	Cells of the...	Specialised to...
Circulatory system	Red blood cells	Transport substances, defend the body, regulate temperature
		Remove waste products and unwanted substances, regulate the water content of the body
Muscular system	Muscle cells	Bring about movement
Nervous system	Neurons	Respond to internal and external stimuli and conditions, carry messages for the body work as a coordinated whole
Respiratory system	Alveoli	Deliver oxygen for respiration and remove waste
Reproductive system	Sperm cells / Egg cells	Bring about fertilisation to produce new offspring
Skeletal system	Cartilage / Bone cells	To bring about movement, support and protect internal structures, produce blood cells, store and release calcium

Electron microscopes

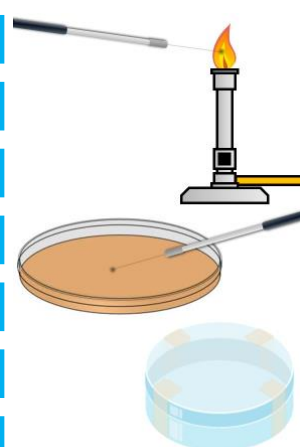
Electron microscopes use a beam of electrons instead of light rays. There are two types of electron microscope:

- The scanning electron microscope (SEM) has a large field of view so can be used to examine the surface structure of specimens. SEMs are often used at lower magnifications.
- The transmission electron microscope (TEM) is used to examine thin slices or sections of cells or tissues.

TEMs have a maximum magnification of around $\times 1\,000\,000$, but images can be enlarged beyond that photographically. The limit of resolution of the transmission electron microscope is now less than 1 nm. The TEM has revealed structures in cells that are not visible with the light microscope.

Growing bacteria

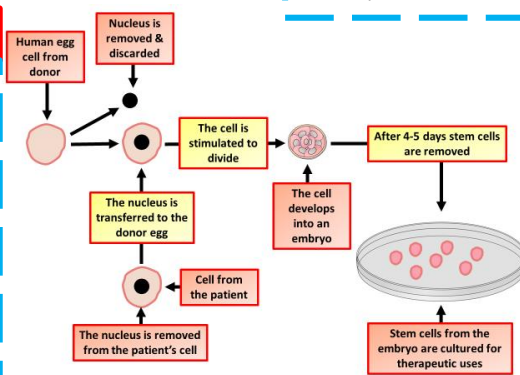
- An inoculating loop can be used to transfer bacteria. It is sterilised by heating it to red hot in a Bunsen flame, before and after use.
- To inoculate the agar, lift the lid of the Petri dish and tilt. Do not fully remove or place on the desk as the lid prevents micro-organisms from the air contaminating the culture, and vice versa
- Following inoculation, the lid of the Petri dish should be secured in place by strips of adhesive tape for safety reasons. The dish should be labelled and dated.
- Inoculated agar plates are incubated at 25°C in school laboratories for no more than 24–48 hours. This encourages growth of the culture without growing human pathogens which thrive at body temperature (37°C).



Therapeutic cloning of stem cells

Therapeutic cloning involves the replication of human embryos to harvest stem cells for medical uses. Most clones are created through a process called "somatic cell nuclear transfer." Essentially, a scientist uses a tiny needle to pull DNA material from the nucleus of a donor cell and transfer it into a hollow egg.

Therapeutic cloning could allow an individual's own cells to be used to treat or cure that person's disease, without risk of introducing foreign cells that may be rejected.



B1 Cell Biology AQA Trilogy

Give three differences between prokaryotes and eukaryotes

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-
-

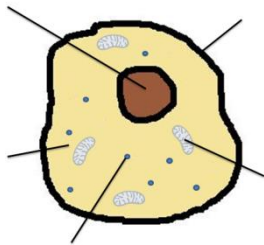
Write the equation used to work out magnification.

Calculate the magnification if you have a magnified image that's 5mm wide and an object that is 0.05mm wide.

Calculate the image size if your specimen is 0.1mm wide and the magnification is x20.



Label the animal cell below



In animals, how are the following cells specialised:

- Sperm cell
- Nerve cell
- Muscle cell

Describe the following stages of the cell cycle

- Interphase
- Mitosis
- Cytoplasmic division

Why is mitosis important?

RPA 1 – Using a light microscope

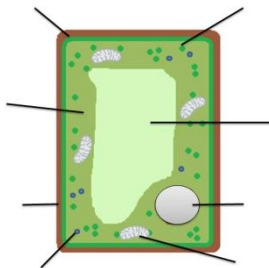


Give 3 features of a good scientific drawing

-
-
-

Name a piece of apparatus that you can use to help measure a specimen

Label the plant cell below



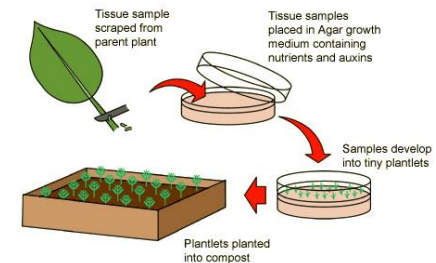
In plants, how are the following cells specialised:

- Root hair cell
- Xylem cell
- Phloem cell

What is a stem cell?

Give a difference between adult human stem cells and plant stem cells (meristem)

Name 2 medical conditions that stem cells could help to treat



What is the function of:

- Mitochondria
- Ribosome
- Nucleus
- Permanent vacuole

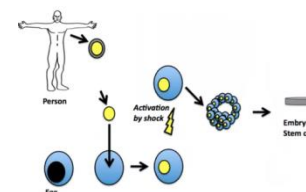
What is a plant cell wall made of?

Define the following terms:
Resolution

Magnification

How has electron microscopy has increased our understanding of cells?

What is therapeutic cloning?



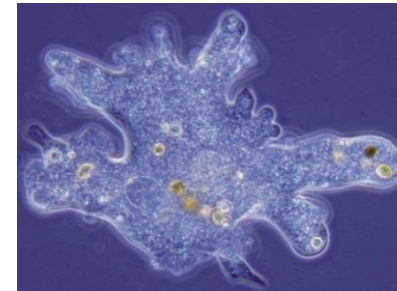
Advantages of producing clones using plant stem cells

Disadvantages of producing clones using plant stem cells

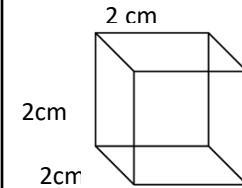
B1 Cell Biology AQA Trilogy

Give a feature of single-celled organisms like amoeba that mean they have sufficient diffusion across the cell membrane to meet the needs of the organism.

Amoeba (single-celled organism)



Calculate the surface area: volume ratio of the cube



Define the term diffusion



Explain why the following features are useful for an exchange surface:

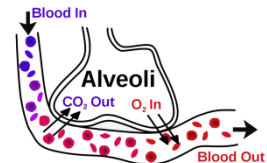
- Thin
- Large surface area
- Good blood supply

Define the term osmosis

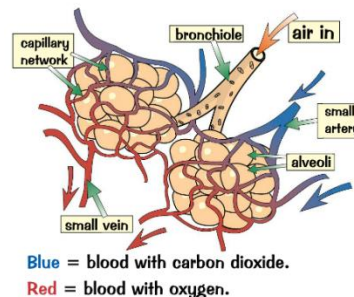
Define the term active transport



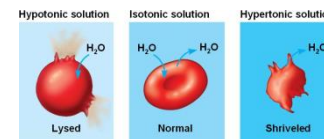
Give examples of diffusion



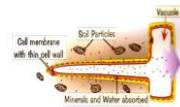
Alveoli in the lungs



Give examples of osmosis



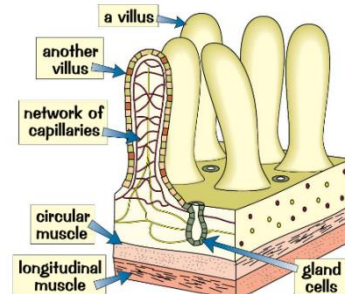
Give examples of active transport



Name 3 factors that affect the rate of diffusion

-
-
-

Villi in the small intestine



Write the formula for calculating percentage change.



RPA 2 – Osmosis potatoes

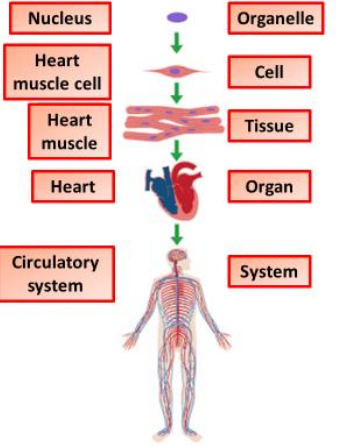
Potato chips were put into different concentrations of salt solution and their change in mass measured.

The start mass of a potato chip was 53.2g and its finish mass was 55.6g. Calculate the percentage change



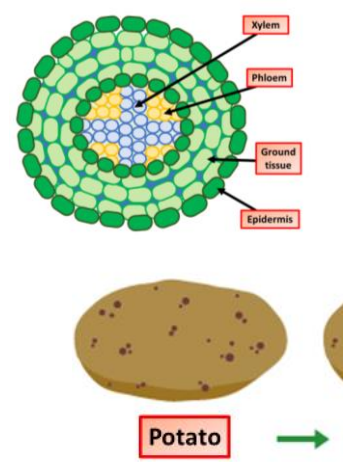
Animal Tissues

In animals, cells are grouped together to form tissues. These tissues vary in their structure, function, and origin. The four important tissues in an animal body are epithelial, connective, muscular and nervous tissues.



Plant Tissues

Plant tissues come in several forms: vascular, epidermal, ground, and meristematic. Each type of tissue consists of different types of cells, has different functions, and is located in different places.



Exchange Surfaces

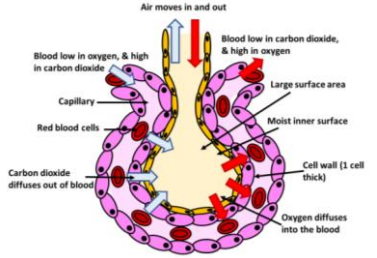
Organisms must take in food, oxygen and water, and other essential substances, from the environment. Plants also need carbon dioxide for photosynthesis. Organisms also need to remove waste substances. Small organisms exchange these essential and waste substances between themselves and the environment. They do this over their body surface. Simple chemical substances can diffuse in and out of their bodies. Inside their bodies, in small organisms, substances don't have to move far. The size of their surface, or surface area, defines how quickly they can absorb substances. The size of their volume defines how much of these substances they need.

Testing for Starch

In the presence of starch, iodine turns a blue/black colour. It is possible to distinguish starch from glucose (and other carbohydrates) using this iodine solution test. For example, if iodine is added to a peeled potato then it will turn black. Benedict's reagent can be used to test for glucose.

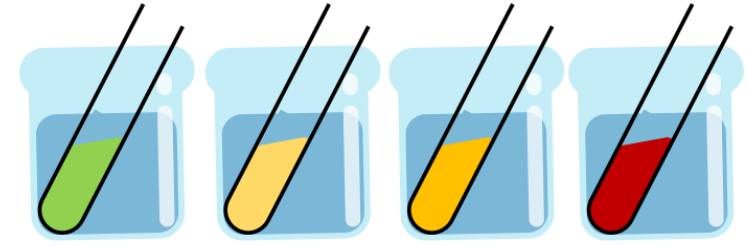
Gaseous Exchange Surfaces

- Key Features**
- Good ventilation
 - Good circulation
 - A large surface area
 - Moist surface
 - Short diffusion distance
 - Concentration gradient
 - Movement of gases



Testing For Sugars

Benedict's solution changes colour when heated



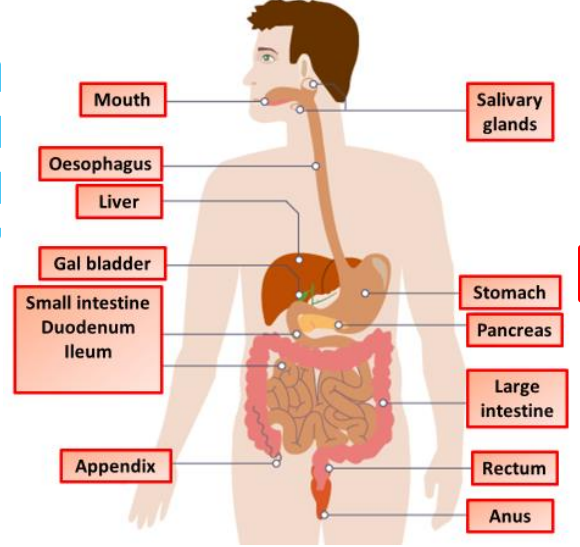
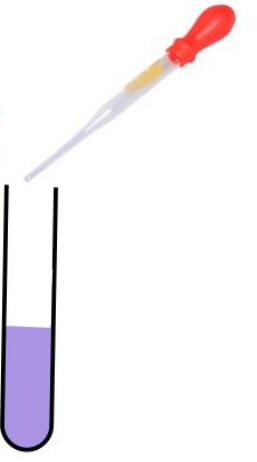
Sugars classed as reducing sugars will react with Benedict's solution on heating for a few minutes. Glucose is an example of a reducing sugar.

Reducing sugars give a red-brown precipitate with Benedict's solution. The precipitate takes a while to settle in the tube – you're more likely to see simply a red or brown colour. If there's not much glucose present, the final colour may be green or yellow, or orange if there's a little more.

Testing for Proteins

- Use a pestle and mortar to grind up a small sample of food.
- Transfer the ground up food into a small beaker. Then add distilled water.
- Stir the mixture so that some of the food dissolves in the water.
- Filter using a funnel with filter paper to obtain as clear a solution as possible.
- The solution should be collected in a conical flask.
- Put 2 cm³ of this solution into a test tube
- Add 2 cm³ of Biuret solution to the solution in the test tube. Shake gently to mix
- Note any colour change. Proteins will turn the solution pink or purple.

A purple colour indicates protein is present

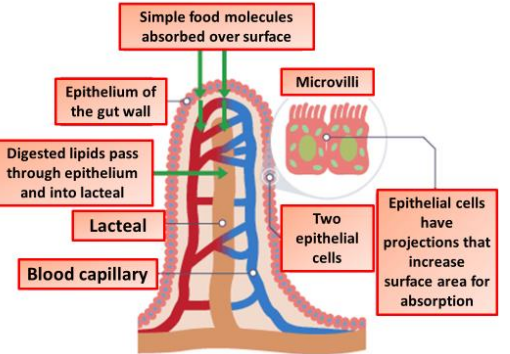


Digestion

The GI tract is a series of hollow organs joined in a long, twisting tube from the mouth to the anus. The hollow organs that make up the GI tract are the mouth, oesophagus, stomach, small intestine, large intestine, and anus. The liver, pancreas, and gallbladder are the solid organs of the digestive system.

Villi

Villi (singular is villus) are small, finger-like structures in the small intestine. They help to absorb digested food. Each villus has micro-villi which increase the surface area of the intestinal walls. A larger surface area allows nutrients to be taken in more quickly.

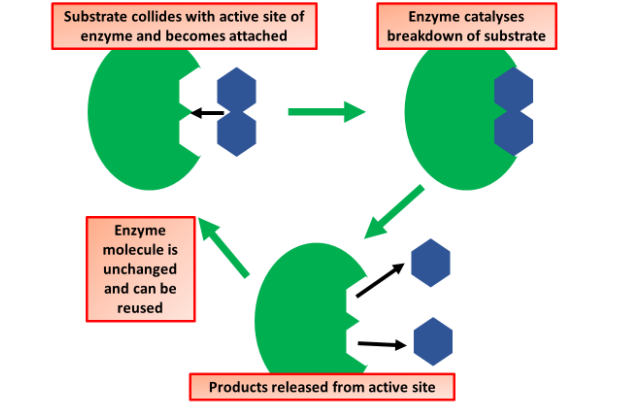


Enzymes

Enzymes are protein molecules in cells which work as catalysts. Enzymes speed up chemical reactions in the body, but do not get used up in the process. Almost all biochemical reactions in living things need enzymes. With an enzyme, chemical reactions go much faster than they would without the enzyme.

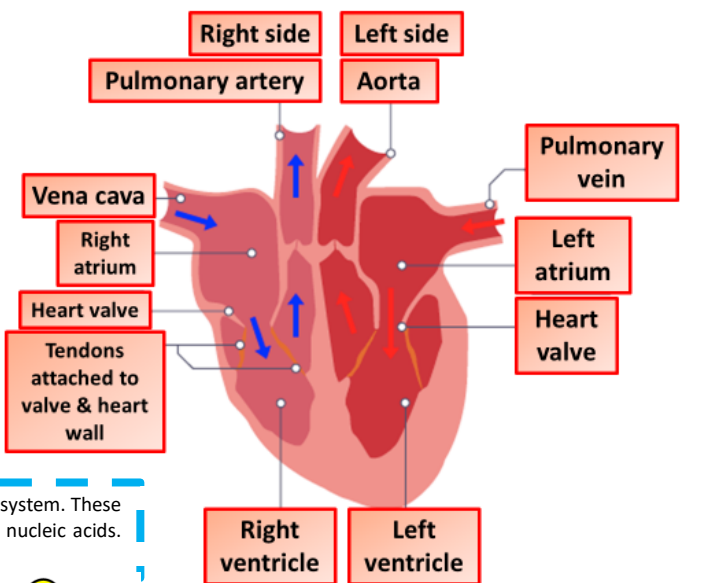
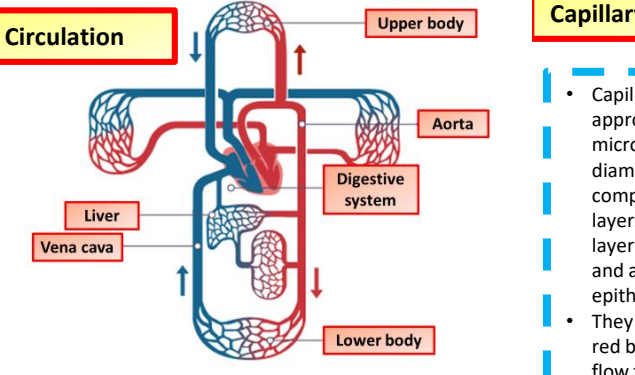
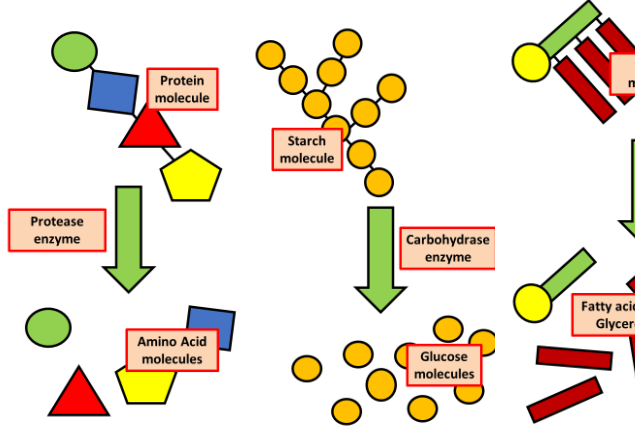
Investigating Enzymes

1. You will investigate the breakdown of **starch** by amylase at different pHs.
2. The different pHs under investigation will be produced using **buffer solutions**. Buffer solutions produce a particular pH, and will maintain it if other substances are added.
3. The amylase will break down the starch.
4. A series of test tubes containing a mixture of starch and amylase is set up at different pHs.
5. A sample is removed from the test tubes every 10 seconds to test for the presence of starch. **Iodine solution** will turn a blue/black colour when starch is present, so when all the starch is broken down, a blue-black colour is no longer produced. The iodine solution will remain orange-brown.
6. For each pH investigated, record the time taken for the disappearance of starch, ie when the iodine solution in the spotting tile remains orange-brown.



Digestive Enzymes

Digestive enzymes are released, or secreted, by the organs of the digestive system. These enzymes include proteases that digest proteins, and nucleases that digest nucleic acids. Examples of digestive enzymes are: Amylase, produced in the mouth.



The Heart

Blood enters the heart through the atria. Blood from two vena cavae enters the right atrium. Blood from the pulmonary veins enters the left atrium. The atria fill, followed by the ventricles. Blood is prevented from flowing back into the atria by heart valves. Blood leaves the heart in the body's main artery - the aorta - from the left side, and the pulmonary artery, from the right.

Artery

- The arteries are the blood vessels that deliver oxygen-rich blood from the heart to the tissues of the body.
- Each artery is a muscular tube lined by smooth tissue and has three layers:
- The intima, the inner layer lined by a smooth tissue called endothelium.

Veins

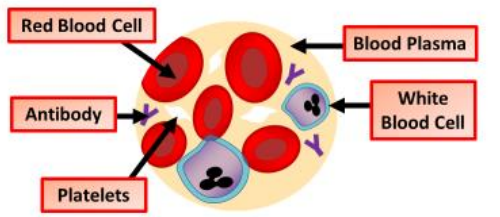
- A vein is an elastic blood vessel that transports blood from various regions of the body to the heart.
- Unlike the high pressure arterial system, the venous system is a low pressure system that relies on muscle contractions to return blood to the heart.

Capillary

- Capillaries are very thin, approximately 5 micrometers in diameter, and are composed of only two layers of cells; an inner layer of endothelial cells and an outer layer of epithelial cells.
- They are so small that red blood cells need to flow through them

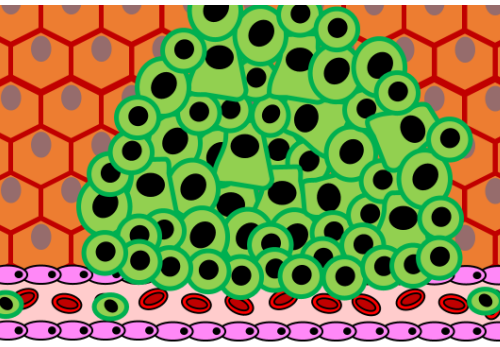
Blood

It has four main components: plasma, red blood cells, white blood cells, and platelets. Blood has many different functions, including: transporting oxygen and nutrients to the lungs and tissues. forming blood clots to prevent excess blood loss.



Cancer

Cancers occur when cell division goes wrong. This causes cells to grow out of control, which form a tumour . There are two types of tumour: malignant - these tumours are cancerous and can break apart, move around the body and start new instances of the same cancers in a process called metastasis.

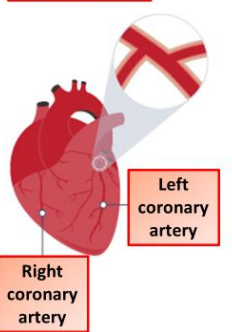


Transpiration is the process by which moisture is carried through plants from roots to small pores on the underside of leaves, where it changes to vapor and is released to the atmosphere. Transpiration is essentially evaporation of water from plant leaves.

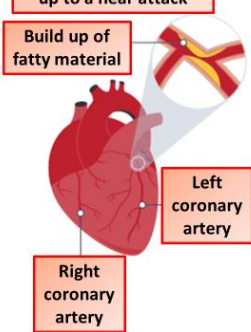
When the plant opens its stomata to let in carbon dioxide, water on the surface of the cells of the spongy mesophyll and palisade mesophyll evaporates and diffuses out of the leaf. This process is called transpiration Water molecules inside the xylem cells are strongly attracted to each other.

The water, warmed by the sun, turns into vapor (evaporates), and passes out through thousands of tiny pores (stomata) mostly on the underside of the leaf surface. This is transpiration. It has two main functions: cooling the plant and pumping water and minerals to the leaves for photosynthesis.

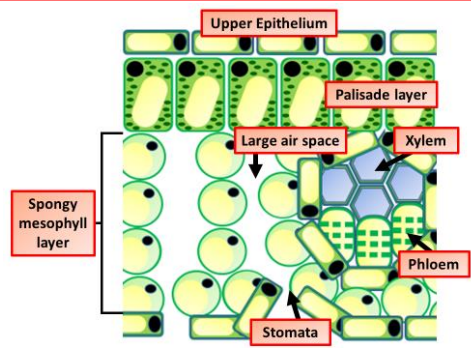
Heart showing a normal artery



Heart showing a coronary artery leading up to a hear attack



Leaf Structure

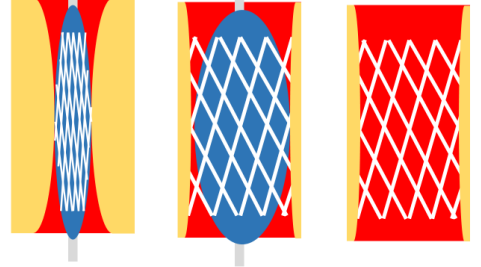


Plants make food in their leaves. The leaves contain a pigment called chlorophyll, which colors the leaves green. Chlorophyll can make food the plant can use from carbon dioxide, water, nutrients, and energy from sunlight. This process is called photosynthesis.

Transpiration

Coronary Heart Disease

The heart is a muscular pump. Like all muscles, it needs oxygen for aerobic respiration to contract. The coronary arteries supply blood, and therefore oxygen, to the heart muscle. The coronary arteries may become blocked by a build-up of fatty material, caused by certain kinds of 'bad' cholesterol. As the fatty material increases, one or more coronary arteries narrow, and can become blocked.

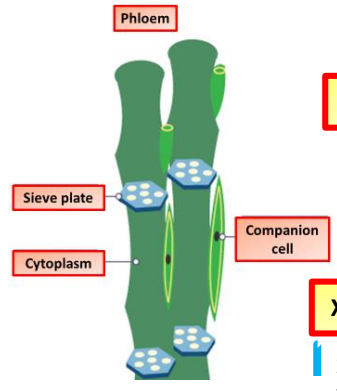


Stents

Coronary arteries that are blocked or have become narrow can be stretched open and have a stent inserted to restore and maintain blood flow. The stent is inserted into a coronary artery in a catheter. Stents are made from metal alloys and do not lead to an immune response in the patient.

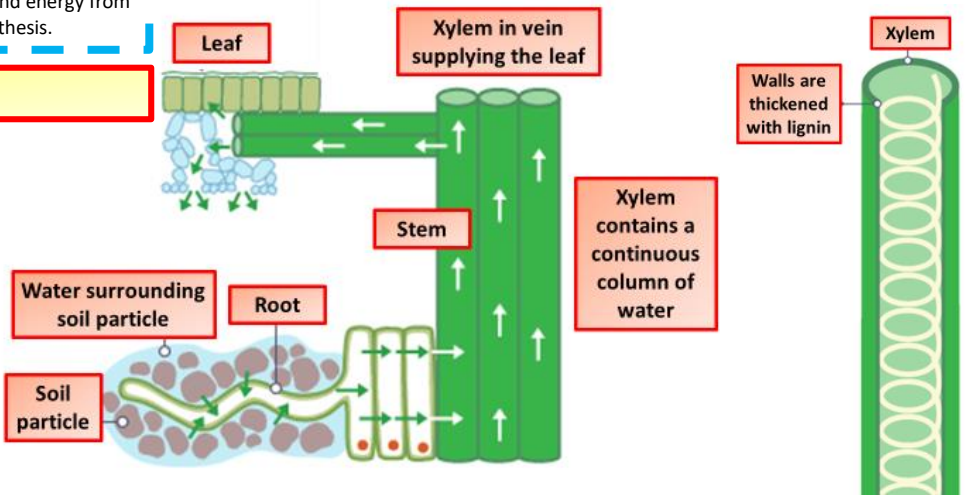
Phloem

Phloem, also called bast, tissues in plants that conduct foods made in the leaves to all other parts of the plant. Phloem is composed of various specialized cells called sieve tubes, companion cells



Xylem

Xylem is one of the two types of transport tissue in vascular plants, phloem being the other. The basic function of xylem is to transport water from roots to stems and leaves, but it also transports nutrients.



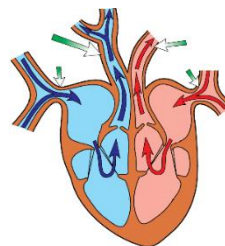
B2 Organisation AQA Trilogy

How do you calculate rate for an enzyme reaction?

At pH6, the time taken for amylase to break down starch was 90s.
Calculate the rate of reaction.



Label the heart diagram



RPA 3: Test for carbohydrates, lipids and proteins

Describe the test for starch

Describe the test for reducing sugars



Define the following terms:
Tissue

Organ

Organ system

Use the 'Lock and key' mechanism to explain enzyme action



Where does the right ventricle pump blood to?

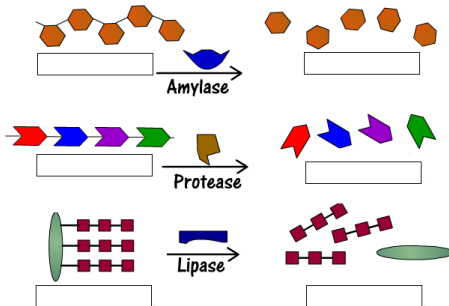
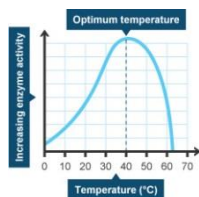
Where does the left ventricle pump blood to?

Why does the left side have thicker muscle than the right side?

Describe the test for protein

Describe the test for lipids

Describe how temperature affects enzyme activity



How is heart rate controlled?

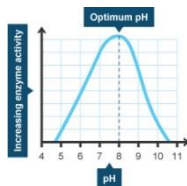
RPA 4: Investigate the effect of pH on the rate of reaction for amylase enzyme

Give a brief account of this investigation

How could you control the temperature during this investigation?



Describe how pH affects enzyme activity



Describe the role of bile in the digestive system

Describe the structure and function of:
Arteries Veins Capillaries



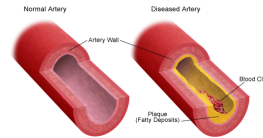
Write the equation for calculating rate of blood flow

1464ml of blood passed through an artery in 4.5 minutes. Calculate the rate of blood flow through the artery in ml/min



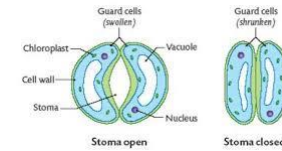
B2 Organisation AQA Trilogy

What is Coronary heart disease?



State risk factors that can increase the incidence of non-communicable disease

Describe the role of stomata and guard cells



Describe the functions of the 4 main components in blood:

Red blood cells

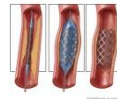
White blood cells

Platelets

Plasma

For each treatment method of CHD give an advantage and a disadvantage
Using a drug (Statins)

Using a stent



Give a risk factor for the following:
Coronary heart disease

Type 2 diabetes

Lung cancer

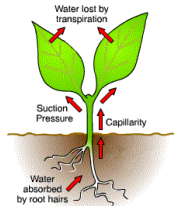
Liver cirrhosis

Low birthweight in babies

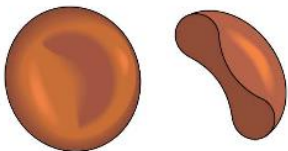
Mental health issues in babies

Cancer

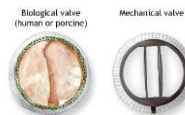
Describe the process of transpiration



How are red blood cells adapted to their function?



What is the importance of valves in the heart and give two ways in which these can be repaired.



What is cancer?

Explain the difference between a benign and malignant tumour

How do the following affect the rate of transpiration?

Temperature

Humidity

Light intensity

Air movement

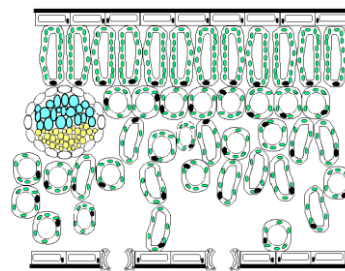
Define the following terms:
Health

Disease

Communicable

Non-communicable

Label the leaf below



Adaptations of plant tissues:

Epidermal tissue

Palisade mesophyll

Spongy mesophyll

Xylem and phloem

Meristem

In plants, how are the following cells specialised:

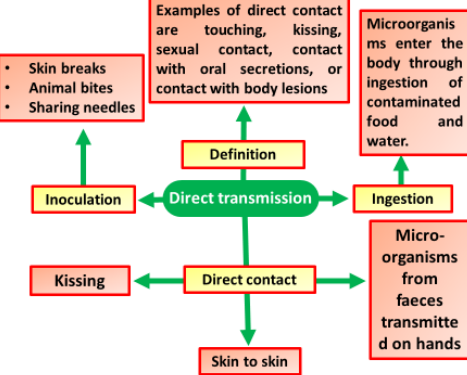
• Root hair cell

• Xylem cell

• Phloem cell

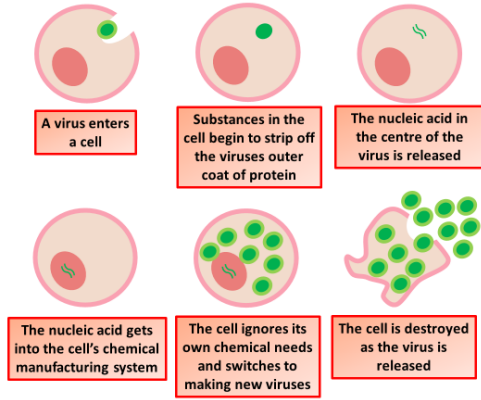
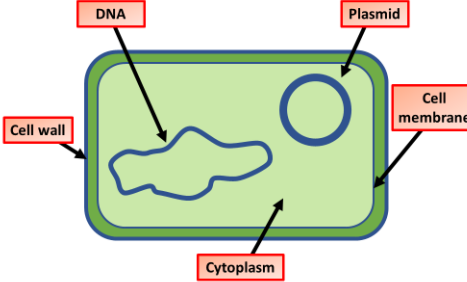
Communicable diseases

All types of pathogen have a simple life cycle. They infect a host, reproduce themselves or replicate if it is a virus, spread from their host and infect other organisms. They also all have structural adaptations that make them successful at completing their life cycles, which enable them to cause further disease.



Pathogens

A pathogen is an organism that causes a disease.. Bacteria are all single-celled. The cells are all prokaryotic . This means they do not have a nucleus or any other structures which are surrounded by membranes. Bacteria can have one or more flagella A virus is a biological agent that reproduces inside the cells of living hosts. When infected by a virus, a host cell is forced to produce thousands of identical copies of the original virus at an extraordinary rate Pathogenic fungi have an enormous impact on human health. Most people are aware of some of the superficial infections caused by fungi. These include skin and nail infections such as athlete's foot and ringworm, predominantly caused by dermatophytes



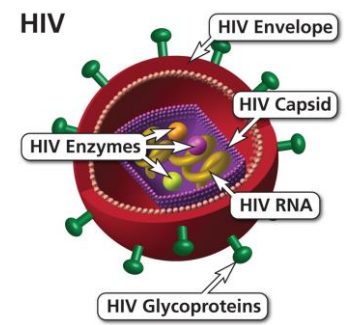
Tobacco Mosaic Virus

The tobacco mosaic virus infects tobacco and lots of other closely related species, such as tomatoes and peppers. It is transmitted by contact between plants, either naturally or through the hands of farmers. It infects the chloroplasts of plant leaves and changes their colour from green to yellow or white in a mosaic pattern. It can also make leaves crinkle or curl up. This reduces the plant's ability to photosynthesise and grow properly, which reduces the crop yield for farmers.



HIV & Aids

HIV stands for human immunodeficiency virus. This infection is transmitted by body fluids, often during unprotected sex, but also through cuts and injecting drugs using shared needles. Immediately after infection, people often suffer mild flu-like symptoms. These pass and for a period of time infected people might not know they are infected. AIDS stands for acquired immune deficiency syndrome. Months or years after the infection of the HIV virus, it becomes active and starts to attack the patient's immune system. HIV at this point has become AIDS.



Gonorrhoea

Gonorrhoea is a sexually transmitted disease (STD) caused by a bacterium. It is a common infection, especially amongst people aged 15-24. Gonorrhoea causes a burning pain when urinating and often forms a thick yellow or green discharge from an infected person's penis or vagina. If untreated it can result in infertility. To prevent infection, people can abstain from having anal, oral or vaginal sex or use a barrier-type of contraception like a condom. Gonorrhoea is treated by antibiotics, though some scientists believe that these antibiotics may one day stop working.

Measles

Measles is a very infectious viral disease that is often caught by young children. It is transmitted through the air in tiny droplets after an infected person sneezes. It causes a fever and skin rash. Many children in developed countries are given vaccines against measles, but sadly this is not the case throughout the world. Infection can cause more serious effects like infertility in adults who did not catch the disease as children.

Salmonella

Salmonella is a genus of bacteria that causes food poisoning. This often means abdominal cramps, vomiting and diarrhoea. It is often found in unhygienic kitchens, undercooked foods such as meat, eggs and poultry, or the same foods that have not been reheated properly. To prevent the spread of salmonella in the UK, all poultry are vaccinated against it. Cooking food thoroughly, after preparing it in hygienic conditions, is the best way to avoid illness.

Rose Black Spot

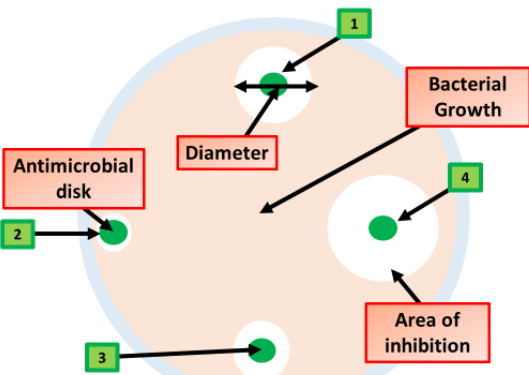
Rose black spot is caused by a fungus which infects roses. It infects leaves and causes black or purple spots on the leaves. The rest of the leaves often turn yellow and can drop off the plant. This reduces that plant's ability to photosynthesise and reduces growth. It can be transmitted in air or water, as well as through direct contact by gardeners. It is treatable using fungicides and by removing and destroying infected leaves.

Athletes Foot

Athlete's foot is a rash caused by a fungus that is usually found between people's toes. It causes dry, red and flaky or white, wet and cracked skin. It is often found in communal areas like swimming pool changing rooms or gyms. It is transmitted by touching infected skin or surfaces that have been previous contaminated. It is treated by antifungal medication.

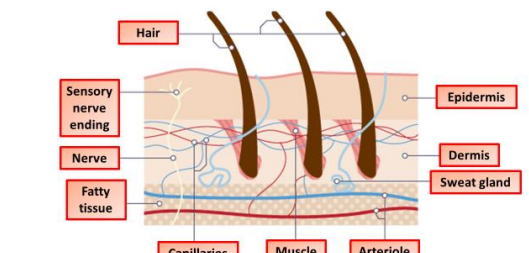
Investigating Pathogens

- 1. Glass Petri dishes and agar gel must be sterilised before use in an autoclave, or pre-sterilised plastic Petri dishes can be bought.
- 2. Pour the sterile agar plates and allow to set fully.
- 3. Sterilise the inoculating loop, by heating it in the Bunsen burner flame.
- 4. Dip the inoculation loop into the microorganism solution and make streaks on the surface of the agar plate.
- 5. Replace the lid as soon as possible, secure with tape. Label and invert the plate, and store upside down.
- 6. Incubate at a maximum temperature of 25°C in schools and colleges.



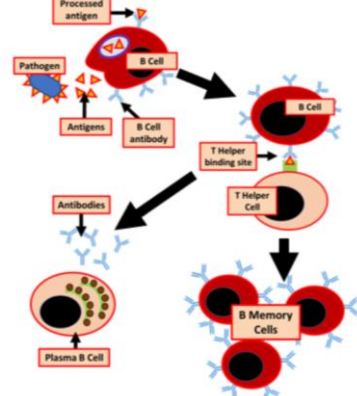
Malaria

The symptoms of malaria include a fever, sweats and chills, headaches, vomiting and diarrhoea. Of the 200 million people infected each year, up to half die from this disease. There is no vaccination for malaria. Infection can only be prevented by stopping individuals from being bitten. People sleep under mosquito nets and wear insect repellent to avoid bites. Antimalarial drugs are also taken, which treat the symptoms and can prevent infection. GM mosquitos are also being introduced which are sterile and so can not reproduce.



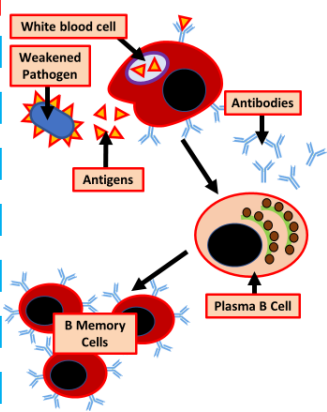
Immune Response

If pathogens pass the non-specific first line of defence they will cause an infection. However, the body has a second line of defence to stop or minimise this infection. This is called the immune system. As a part of this there are two types of white blood cell called phagocytes and lymphocytes.



Vaccination

Pathogens are microbes that cause diseases. Vaccines allow a dead or altered form of the disease causing pathogen to be introduced into the body, which contain a specific antigen. This causes the immune system, specifically the white blood cells, to produce complementary antibodies, which target and attach to the antigen. When a white blood cell engulfs and digests a pathogen it is called phagocytosis.



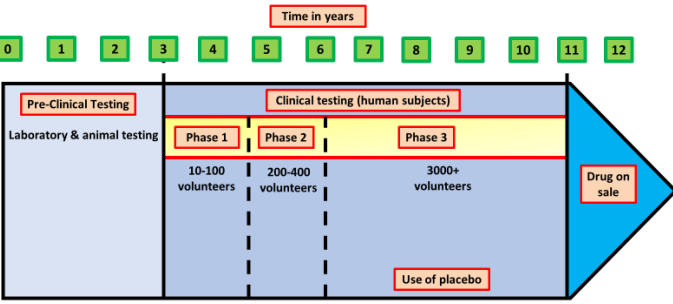
Antibiotics

Antibiotics are substances that slow down or stop the growth of bacteria. They are commonly prescribed medicines, examples include penicillin and amoxicillin. These can be taken to cure the disease by killing the pathogen, but only cure bacterial diseases and not viral ones.

Physical Barriers to Infection

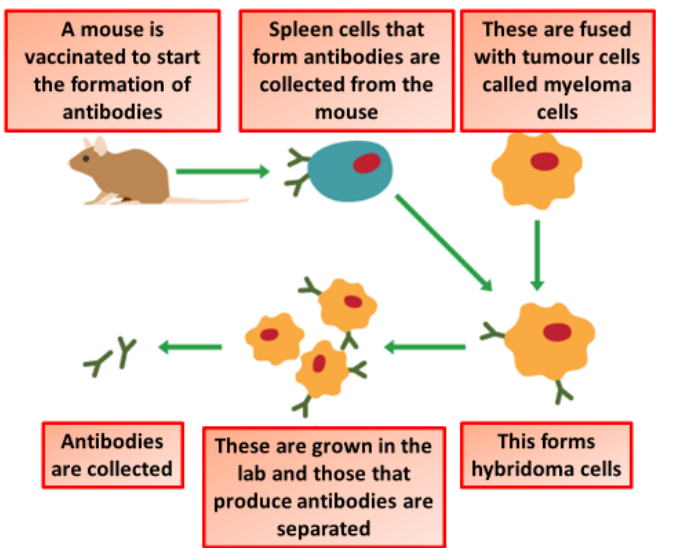
The body is constantly defending against attacks from pathogens. The first line of defence against infection stops the pathogens from entering your body. These first lines are general defences, and are not specific to fight against certain types of pathogen. They are called non-specific, and they can be physical or chemical barriers. Examples include: Skin, lysozyme in tears, Mucus, stomach acid.

Drug Development



Monoclonal Antibodies

Antibodies bind to specific antigens on pathogens. This means that only one type of antibody will bind to a matching antigen. Scientists discovered that we could make antibodies to bind to antigens on other substances, and not just those on pathogens. Once bound, the antigens - and the substances they are found on - are merged tightly together. This makes them easier to identify and deal with



Antibiotic Resistance

Bacteria can evolve quickly because they reproduce at a fast rate. Mutations in the DNA of bacteria can produce new characteristics. A random mutation might cause some bacteria to become resistant to certain antibiotics, such as penicillin. Antibiotics usually kill bacteria, but in this case the mutation means the bacteria cannot be destroyed by the antibiotic. This evolution of antibiotic resistance in bacteria is an example of natural selection leading to evolution. The bacteria can evolve so quickly because they reproduce rapidly.

B3 Infection and response

AQA Trilogy

TMV (tobacco mosaic virus)
Which plant does it affect?

Symptoms

Problem it causes to the plant

Malaria
Pathogen caused by



Symptoms

Spread by

How can the spread be controlled?

Describe what is in a vaccination and how it works



How can the spread of disease be reduced or prevented?

Salmonella
Pathogen caused by

Symptoms

Spread by



Describe how the following non-specific defences help to prevent the entry of pathogens
Skin

Nose

Trachea and bronchi

Stomach

What is an antibiotic used for and give an example

What can antibiotics not kill?

What is the role of a painkiller?

Measles
Pathogen caused by

Symptoms

Spread by

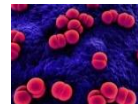


Gonorrhoea
Pathogen caused by

Symptoms

Spread by

How can the spread be controlled?



Describe the process of phagocytosis



Where did the following drugs originate from?

Heart drug digitalis

Painkiller aspirin

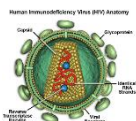
Penicillin antibiotic

HIV
Pathogen caused by

Symptoms

Spread by

What is AIDs?



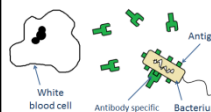
Rose black spot
Pathogen caused by

Symptoms

Spread by

How can it be treated?

Describe the role of:
Antibodies



Antitoxins

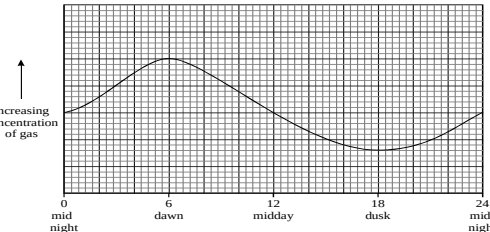


Define the following terms
Toxicity

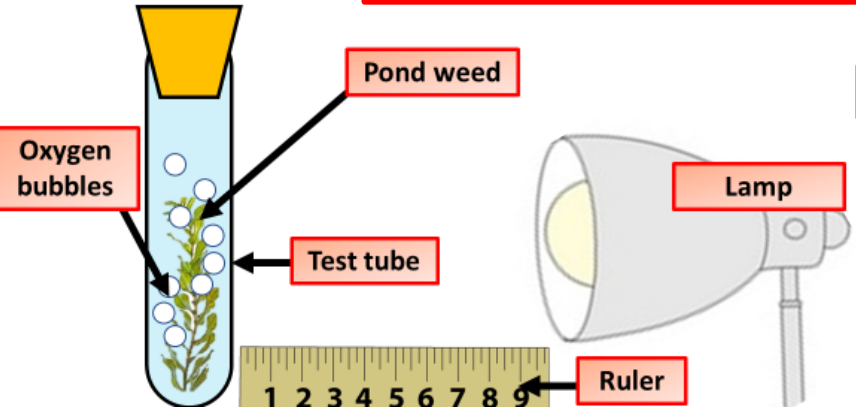
Efficacy

dose

Photosynthesis



There are three main factors light, intensity carbon dioxide concentration and also temperature these factors are called limiting factors light is essential for photosynthesis as it provides the energy to split the water and therefore enable carbon dioxide and water to react no light, no photosynthesis increasing

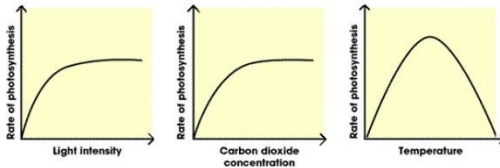


Method

1. Set up a boiling tube containing 45 cm³ of sodium hydrogencarbonate solution (1%). Allow the tube to stand for a few minutes and shake to disperse any air bubbles that might form.
2. Cut a piece of the pondweed, *Cabomba*. The pondweed should be 8 cm long.
3. Use forceps to place the pondweed in the boiling tube carefully. Make sure that you don't damage the pondweed, or cause the liquid to overflow.
4. Position the boiling tube so that the pondweed is 10 cm away from the light source. Allow the boiling tube to stand for five minutes. Count the number of bubbles emerging from the cut end of the stems in one minute. Repeat the count five times and record your results.
5. Calculate the average number of bubbles produced per minute. Repeat the experiment at different distances away from the light source.

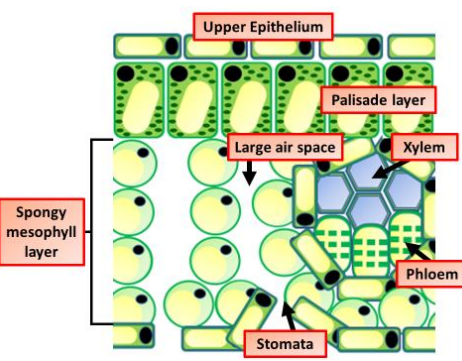
Plants produce glucose by a process called photosynthesis. The plant uses glucose to grow. The graph shows the change in concentration of carbon dioxide in a glasshouse full of plants over 24 hours. Describe what the plant needs for photosynthesis, where these things come from and why the concentration of carbon dioxide changes over 24 hours. Describe what the plant needs for photosynthesis, where these things come from and why the concentration of carbon dioxide changes over 24 hours.

Limiting Factors



Investigating Photosynthesis

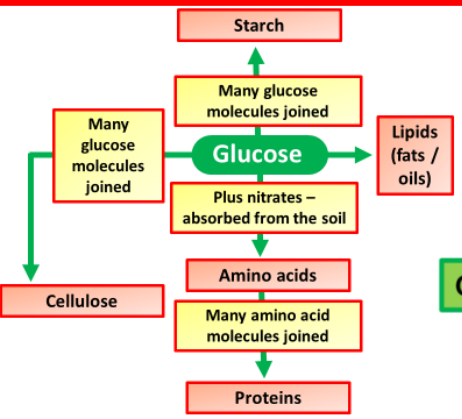
Leaf structure



Method

1. The leaf of is dropped in boiling water to stop any further reactions occurring.
2. The leaf is left for 10 minutes in hot ethanol in a boiling tube. This removes the chlorophyll
3. The leaf is dipped in boiling water to soften it
4. The leaf is spread out in a Petri dish and covered with iodine solution
5. The areas that had the chlorophyll stain blue-black. The areas that had no chlorophyll remain pale

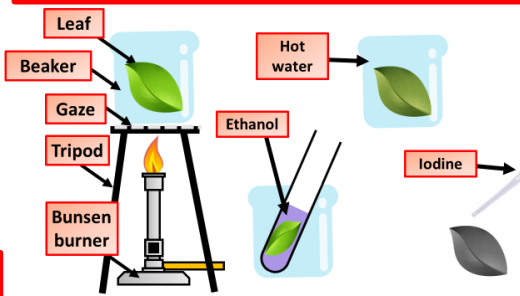
Uses of Glucose



Metabolism is the term used for all the chemical reactions that go on inside an organism's body. These reactions build up molecules, and break them down. They are controlled by enzymes.

Waxy cuticle: See through layer which lets light through and stops the leaf from becoming water logged.
Palisade layer: Cells have lots of chloroplasts so can carry out lots of photosynthesis.
Spongy mesophyll layer: Rounded cells which, creates large air spaces within the leaf.
Guard cells: Open and close the air holes on the underside of the leaf so control the rate of transpiration.
Air spaces: Enable movement of carbon-dioxide into the leaf and oxygen out.
Phloem: Carries away glucose
Xylem: Gives a good supply of water to the leaf so increases the rate of photosynthesis

Testing leaves for starch



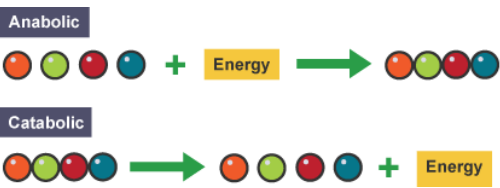
Aerobic Respiration



Anaerobic Respiration

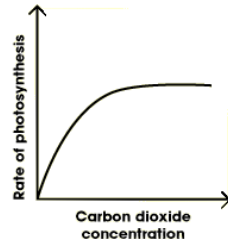


Metabolism



B4 Bioenergetics AQA Trilogy

How does carbon dioxide concentration affect the rate of photosynthesis?



Write the word and symbol equations for aerobic respiration

Is this exothermic or endothermic?

RPA 5: Investigate the effect of light intensity on the rate of photosynthesis



State the gas given off by plants during photosynthesis

Which piece of apparatus would you use to find the volume of gas given off?

Describe how you could control the effect of heat from the lamp on the rate of photosynthesis

Write the word AND symbol equation for photosynthesis

Is it exothermic or endothermic?

What is meant by the term limiting factor?

Give three reasons that living organisms need energy

-
-
-

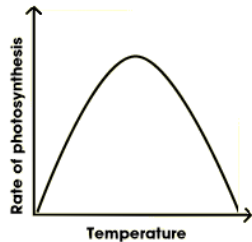


Why do heart rate and breathing rate increase during exercise?

What is meant by the term oxygen debt?



How does temperature affect the rate of photosynthesis?



Inverse square law

This is the 'proportional to' symbol. Putting one over the distance shows the inverse

$\text{light intensity} \propto \frac{1}{\text{distance (d)}^2}$

The distance is squared.

Calculate the light intensity when the lamp is 10cm from the pondweed in a.u

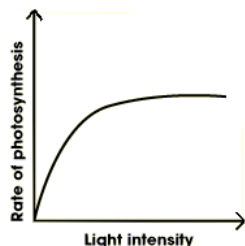


Write the word equation for anaerobic respiration in muscles

How does the energy given out compare to aerobic respiration?

Define the term metabolism

How does light intensity affect the rate of photosynthesis?



List 5 uses of the glucose produced during photosynthesis

-
-
-
-
-

Write the word equation for anaerobic respiration in plant and yeast cells



In yeast, this is also known as fermentation. Give 2 uses of this.

What does metabolism include?

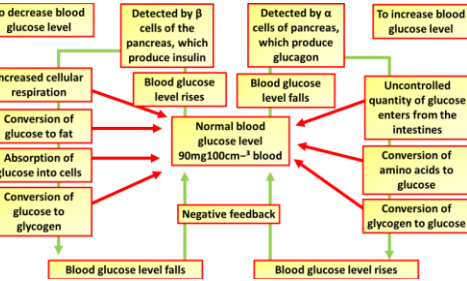
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Homeostasis

Homeostasis is the maintenance of a constant internal environment. The nervous system and hormones are responsible for this. One example of homeostasis is the concentration of carbon dioxide in the blood being carefully controlled.

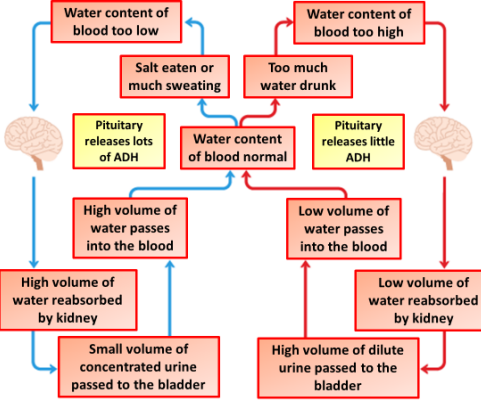
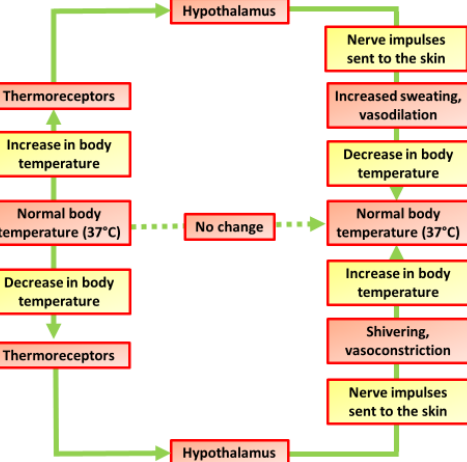
Controlling Blood Glucose

Insulin and glucagon are hormones secreted by islet cells within the pancreas. They are both secreted in response to blood sugar levels, but in opposite fashion! Insulin is normally secreted by the beta cells (a type of islet cell) of the pancreas



Controlling Body Temperature

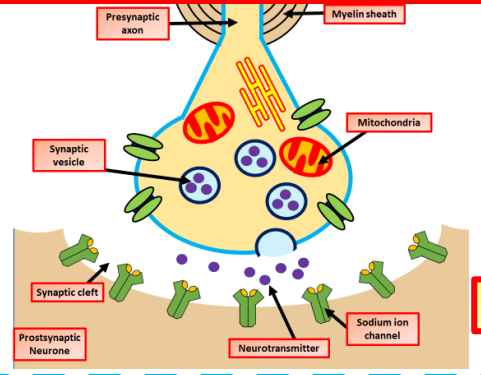
The blood vessels, hairs, and sweat glands of the skin work together to help control body temperature. In addition, tiny muscles attached to the hair follicles pull the hairs erect, trapping an insulating layer of air next to the skin.



Controlling Blood Water Levels

Water levels and mineral salts in the blood are controlled to protect animal cells by stopping too much water from entering or leaving them. The concentration of water and salts is the same inside and outside of the cells. If body cells lose or gain too much water by osmosis, they do not function efficiently.

The Synapse

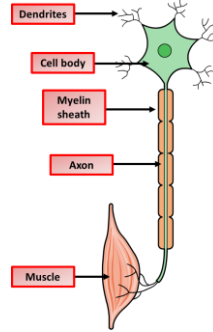


1. An electrical impulse travels along an axon.
2. This triggers the nerve-ending of a neuron to release chemical messengers called neurotransmitters.
3. These chemicals diffuse across the synapse (the gap) and bind with receptor molecules on the membrane of the next neuron.
4. The receptor molecules on the second neuron bind only to the specific chemicals released from the first neuron. This stimulates the second neuron to transmit the electrical impulse

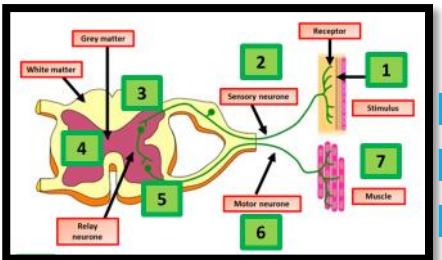
Nervous System

Neurons are nerve cells that carry information as tiny electrical signals. There are three different types of neurones, each with a slightly different function:

- **sensory** neurones carry signals from receptors to the spinal cord and brain
- **relay** neurones carry messages from one part of the CNS to another
- **motor** neurones carry signals from the CNS to effectors.



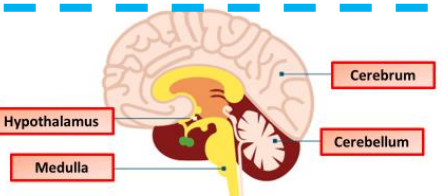
Reflex arc



The nerve pathway is called an arc, the receptor detects a stimulus, sensory neurones send an impulse to relay neurones. These connect with motor neurones which send electrical impulses to an effector. That produces a response such as a muscle contraction.

The Brain

The cerebrum is the large folded area of the brain and is responsible for conscious thoughts, reasoning, memory and emotions. The cerebellum is found at the rear of the brain below the cerebrum and controls balance and coordinated movement.



Investigating reaction time

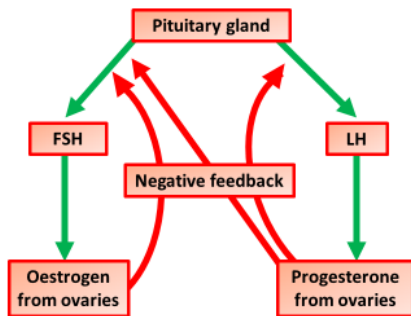
- Work with a partner.
- Person A holds out their hand with a gap between their thumb and first finger.
- Person B holds the ruler with the zero at the top of person A's thumb
- Person B drops the ruler without telling Person A and they must catch it.
- The number level with the top of person A's thumb is recorded in a suitable table. Repeat this ten times.
- Swap places, and record another ten attempts.
- You can use the conversion table to help convert your ruler measurements into reaction time or just record the catch distance in cm.

Negative Feedback (higher)

A negative feedback mechanism is an important type of control that is found in homeostasis. It responds when conditions change from the ideal or set point and returns conditions to this set point. There is a continuous cycle of events in negative feedback.

- In general this works in the following way:
- if the level of something rises, control systems reduce it again
- if the level of something falls, control systems raise it again

The Mensural Cycle



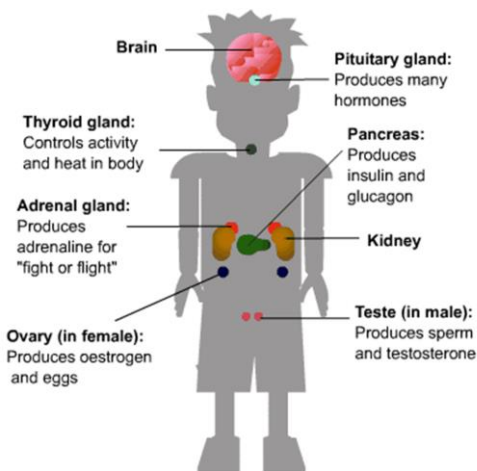
1. The pituitary gland releases FSH
2. FSH moves around in the blood
3. FSH travels to the ovaries
4. FSH causes an egg follicle to mature
5. The developing egg releases oestrogen
6. Oestrogen travels around in the blood
7. Oestrogen causes the uterus lining to thicken
8. When Oestrogen reaches the pituitary gland, it turns off the production of FSH
9. Oestrogen turns on the production of LH
10. LH travels around in the blood
11. LH reaches the ovaries and causes the egg to be released from the follicle
12. The empty egg follicle releases Progesterone
13. Progesterone travels to the pituitary gland and turns off LH production
14. Progesterone maintains the uterus lining for 7 days
15. When the Progesterone runs out the uterus lining breaks down and the woman has a period

Investigating reaction time

- Work with a partner.
- Person A holds out their hand with a gap between their thumb and first finger.
- Person B holds the ruler with the zero at the top of person A's thumb
- Person B drops the ruler without telling Person A and they must catch it.
- The number level with the top of person A's thumb is recorded in a suitable table. Repeat this ten times.
- Swap places, and record another ten attempts.
- You can use the conversion table to help convert your ruler measurements into reaction time or just record the catch distance in cm.

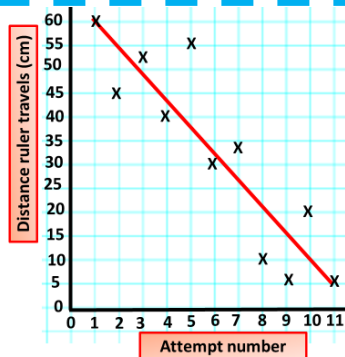
Contraception

Barrier methods, such as condoms, are a form of contraception that help to protect against both sexually transmitted infections (STIs) and pregnancy. You should use condoms to protect both your sexual health and that of your partner, no matter what other contraception you're using to prevent pregnancy.



Endocrine System

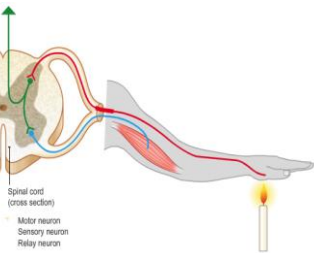
The endocrine system is a chemical messenger system consisting of hormones, the group of glands of an organism that carry those hormones directly into the circulatory system to be carried towards distant target organs, and the feedback loops of homeostasis that the hormones drive.



B5 Homeostasis and response

AQA Trilogy

Draw a flow chart to show the action of the reflex arc below.



Why is the reflex arc important?

Name the 3 types of neurone in the reflex arc

What is a synapse and how does it work?

RPA 6: Reaction times

Explain why using a computer to measure reaction times is better than the ruler drop method.

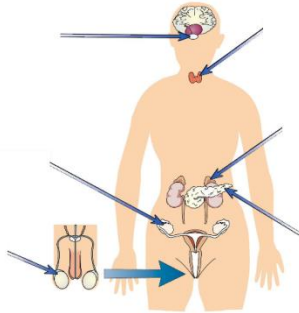


Define homeostasis

List 3 things that need to be kept in homeostasis in the human body

-
-
-

Complete the labels



What is the endocrine system and how does it work?

Describe the role of the pituitary gland

Where is blood glucose monitored?

Describe the two roles of insulin in lowering blood glucose levels

-
-

Describe what occurs if blood glucose levels are too low

KW: glucagon, glycogen, pancreas

Explain how glucagon interacts with insulin in a negative feedback cycle to control blood glucose levels

Compare type 1 and type 2 diabetes

What is meant the term secondary sex characteristics and what causes these to occur?

Where are the following hormones produced and what are their functions?

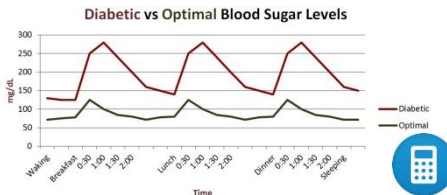
Oestrogen

Testosterone

List the 4 hormones involved in the menstrual cycle and give their functions

-
-
-
-

Describe the effect of insulin on people with and without diabetes



B5 Homeostasis and response

AQA Trilogy

Explain how oral contraceptives work

Explain how spermicidal agents work to prevent pregnancy

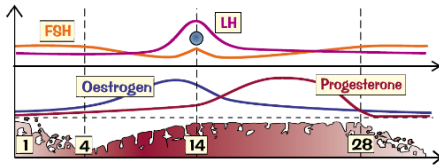
Explain the roles of Thyroxine

Explain how hormones interact to control the menstrual cycle

Explain how Barrier methods such as condoms and diaphragms work

Describe the steps involved in IVF treatment

Explain the role of Adrenaline



What do the following acronyms stand for
IVF

Explain how IUDs prevent pregnancy

List three negative impacts of fertility treatment on a couple

Explain how the following contraceptives work:
Injection

FSH

-

skin patch

LH

-

-

What is an implant and how does it prevent pregnancy?

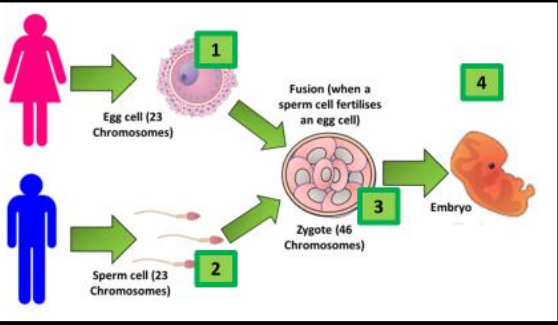
What does abstaining mean?

Give examples of surgical methods of contraception

Evaluate the methods of contraception.

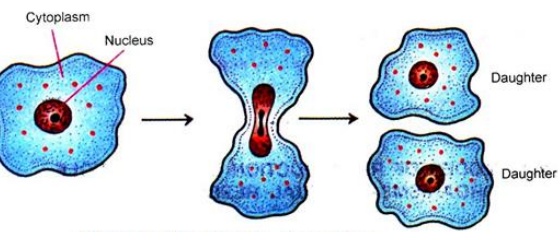
Sexual Reproduction

During sexual reproduction the two gametes join together in a fusion process known as fertilization, to create a zygote, which is the precursor to an embryo offspring, taking half of its DNA from each of its parents



Asexual Reproduction

Asexual reproduction is a type of reproduction by which offspring arise from a single organism, and inherit the genes of that parent only; it does not involve the fusion of gametes, and almost never changes the number of chromosomes.



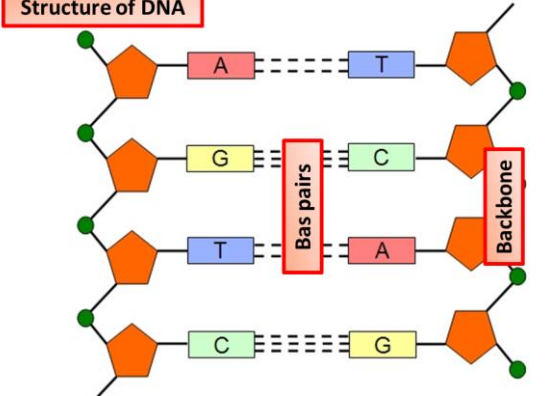
Binary fission in Amoeba

Meiosis

Meiosis is a type of cell division that reduces the number of chromosomes in the parent cell by half and produces four gamete cells. This process is required to produce egg and sperm cells for sexual reproduction.

- Stage 1:** Chromosomes become shorter and thicker
- Stage 2:** Homologous chromosomes pair up in the centre of the cell
- Stage 3:** Genetic crossing over
- Stage 4:** Spindle fibres separate the chromosomes
- Stage 5:** The cell divides into two daughter cells
- Stage 6:** Chromosomes align in the centre of each cell
- Stage 7:** Spindle fibres separate the chromosomes into chromatids
- Stage 8:** Each cell divides into two daughter cells

Structure of DNA



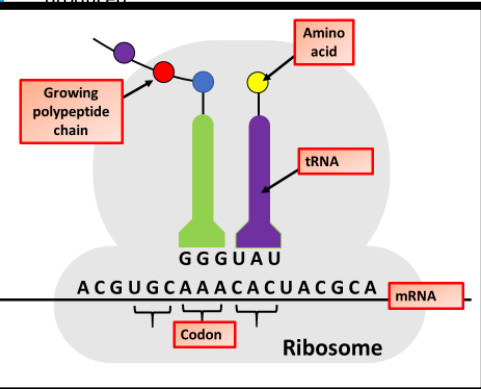
DNA

- A double helix
- DNA shown as a double twisted string with interlinking parts
- Base pairs on a DNA molecule
- Each strand of DNA is made of chemicals called bases. Do not confuse these with the bases you meet when you study acids and alkalis in chemistry. There are four different types of bases, shown as A, T, C and G in the diagram.
- In DNA, two strands coil together to form a double helix. There are chemical cross-links between the two strands, formed by pairs of bases

Protein Synthesis

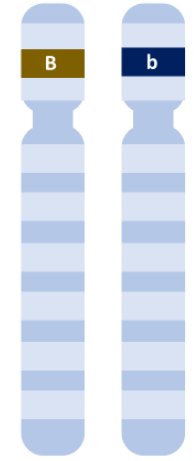
Cells express their genes by converting the genetic message into protein. This process of protein synthesis occurs in two stages - transcription and translation.

- Copy of one side of DNA strand is made (called mRNA, messenger RNA)
- mRNA moves to cytoplasm, then ribosome.
- mRNA goes through ribosome 3 bases at a time.
- transfer RNA (tRNA) matches up with the open DNA bases.
- tRNA releases the amino acid at the top, which joins the chain of amino acids being produced



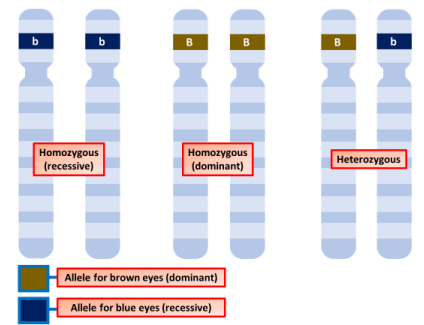
Chromosomes

- The chromosomes of a cell are in the cell nucleus.
- They carry the genetic information.
- Each chromosome contains many genes.
- Chromosomes carry genetic information in a molecule called DNA.
- Each human body cell contains 46 chromosomes.
- These can be arranged into 23 pairs.
- Each chromosome in a pair carries the same types of genes.
- Chromosomes are important to this process to ensure the DNA is accurately replicated.
- 1 of each chromosome comes from each parent



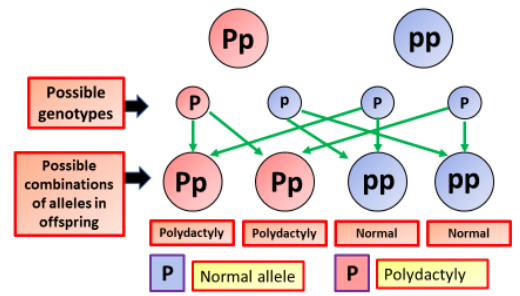
Genetic Inheritance

- A dominant allele is always expressed, even if one copy is present. Dominant alleles are represented by a capital letter, for example you could use a B. The allele for brown eyes, B, is dominant. You only need one copy of this allele to have brown eyes. Two copies will still give you brown eyes.
- A recessive allele is only expressed if the individual has two copies and does not have the dominant allele of that gene. Recessive alleles are represented by a lower case letter, for example, b. The allele for blue eyes, b, is recessive. You need two copies of this allele to have blue eyes.



Genetic disorders

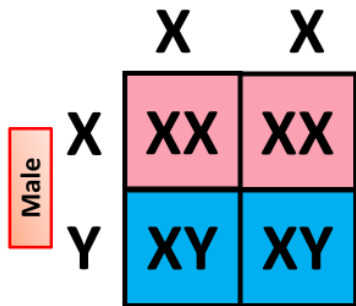
A genetic disorder is a genetic problem caused by one or more abnormalities formed in the genome. Genetic disorders may be hereditary, meaning that they are passed down from the parents' genes. In other genetic disorders, defects may be caused by new mutations or changes to



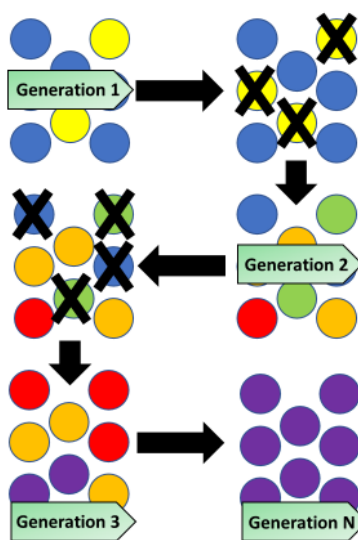
Sex Determination

- In males the sex chromosomes are heterozygous – one X and one Y chromosome.
- In females the sex chromosomes are homozygous – two X chromosomes.

Female



Adult cell cloning



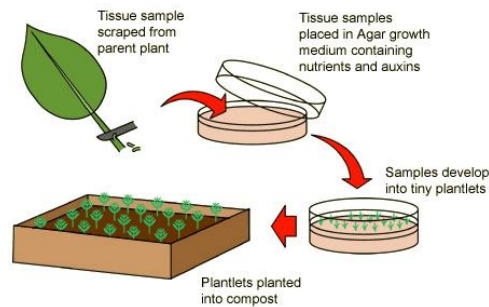
Evolution

- Individuals in a species show a wide range of variation and this variation is because of differences in their genes.
- Individuals with characteristics most suited to their environment are more likely to survive and reproduce. The genes that allow these individuals to be successful are passed to their offspring.
- Those that are poorly adapted to their environment are less likely to survive and reproduce. This means that their genes are less likely to be passed on to the next generation.
- Given time, a species will gradually evolve.

Genetic Engineering

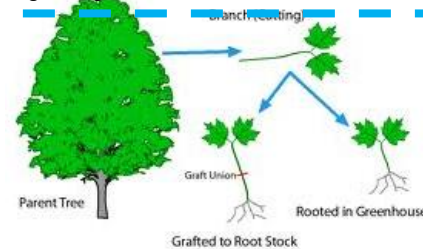
- The cell that you want to extract the gene out of is identified
- The chromosome (DNA) is extracted from the nucleus of that cell
- The specific gene is identified on the chromosome and cut out using restriction enzymes
- Bacteria is used to produce insulin
- The plasmid from a bacteria cell is extracted
- The plasmid is cut open using a restriction enzyme
- The gene for insulin is stuck into the plasmid using an enzyme
- The plasmid is re-inserted back into the bacteria
- The bacterium now has the gene to produce human insulin
- The bacterium is grown in a fermenter
- At pH7, 35°C & with the correct levels of nutrients and oxygen
- As the bacteria replicate they produce insulin

Cloning Plants



Tissue culture

Another way of cloning plants is by tissue culture, which works not with cuttings but with tiny pieces from the parent plant. Sterile agar jelly with plant hormones and lots of nutrients is needed. This makes tissue culture more expensive and difficult to do than taking cuttings.



Cuttings

The simplest way to clone a plant involves taking a cutting. A branch from the parent plant is cut off, its lower leaves removed and the stem planted in damp compost. Plant hormones are often used to encourage new roots to develop. The cutting is usually covered in a clear plastic bag at this stage to keep it moist and warm. After a few weeks, new roots develop and a new plant is produced. The method is easy enough for most gardeners to do successfully.

Embryo Transplant

A developing embryo is removed from a pregnant animal at an early stage, before the embryo's cells have had time to become specialised. The cells are separated, grown for a while in a laboratory, and then transplanted into host mothers. When the *offspring* are born, they are identical to each other. They are not identical to their host mothers, because they contain different genetic information (the offsprings' DNA comes from the original pregnant animal and the father).

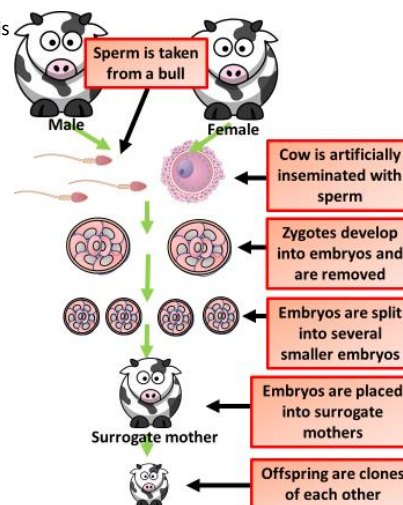
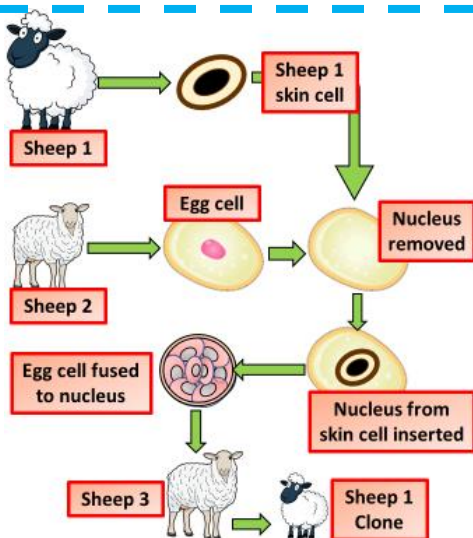
Extinction & Evidence of Evolution

Extinction occurs when there are no remaining individuals of a species alive.

Evidence of evolution

Species may share similar physical features because the feature was present in a common ancestor (homologous structures).

- Molecular biology. DNA and the genetic code reflect the shared ancestry of life.
- Biogeography.
- Fossils.
- Direct observation.



B6 Inheritance variation and evolution AQA Trilogy

Define the following terms:

Gene

Chromosome

Genome

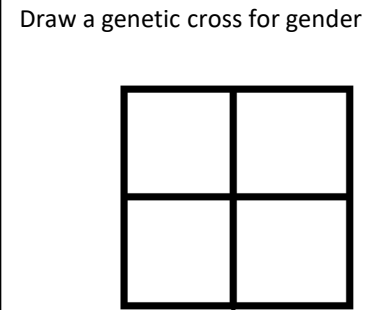
Allele

Define the following terms:

Gamete

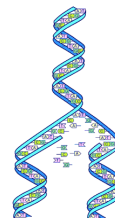
Geneotype

Phenotype



Describe the difference between sexual reproduction and asexual reproduction

Discuss the importance of understanding the human genome



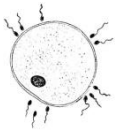
Give examples of characteristics controlled by a single gene

How many chromosomes are in a human body cell?

How many chromosomes are in an egg or sperm cell?

What are the sex chromosomes for:
Male
Female

Describe the process of meiosis and explain why it is important for sexual reproduction



Explain the differences between how cystic fibrosis and polydactyly are passed on to offspring.



Define the following terms:

Dominant

Recessive

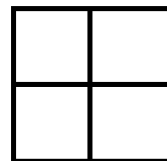
Homozygous

Heterozygous

Differences in the characteristics of individuals in a population is called variation. Describe what causes variation.

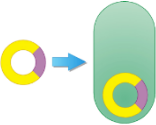
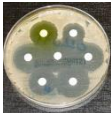
Describe the structure of DNA

Draw a Punnett square to work out the probability that a child will inherit polydactyly if the parents have the genotypes Pp and pp



What are the issues surrounding embryo screening?

What is meant by the term mutation

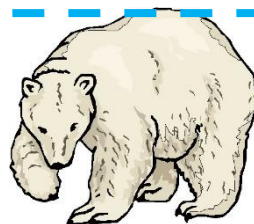
B6 Inheritance variation and evolution AQA Trilogy	Describe genetic engineering 	Describe the process of selective breeding (artificial selection)	Explain how bacteria can become resistant to certain antibiotics 
Describe what is meant by the terms: Evolution Natural selection	Give uses of genetic engineering	Give 4 examples of characteristics that may be selected for • • • •	How can we reduce the number of bacteria becoming resistant to certain antibiotics
Describe how a new species can be formed	Outline some of the pros and cons of GM crops	Give some of the negative impacts of selective breeding	Complete for the organisation of organisms D K P C O F G S
Give 3 ways that fossils are formed. • • •	Describe factors that may contribute to the extinction of a species	Imagine you are a farmer, describe how you would selectively breed cows to produce high volumes of milk  	Why do scientists use the binomial naming system? Give the 3 domains developed by Carl Woese • • •

Adaptations

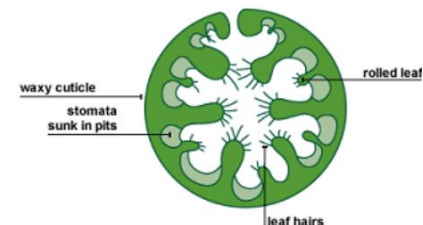


- Large, flat feet to spread their weight on the sand.
- Thick fur on the top of the body for shade, and thin fur elsewhere to allow easy heat loss.
- A large surface area to volume ratio to maximise heat loss.
- The ability to go for a long time without water (they don't store water in their humps, but they lose very little through urination and sweating).
- The ability to tolerate body temperatures up to 42°C.
- Slit-like nostrils and two rows of eyelashes to help keep the sand out.

- a white appearance, as camouflage from prey on the snow and ice
- thick layers of fat and fur, for insulation against the cold
- a small surface area to volume ratio, to minimise heat loss
- a greasy coat, which sheds water after swimming.



Cacti also have spines (the prickly thorn like things) in which replace leaves. These spines minimise the surface area of the cactus reducing water loss. The cacti's stem is also thick and fleshy allowing it to store a lot of water. The stem also has a waxy waterproof kind of coating to help retain more water.



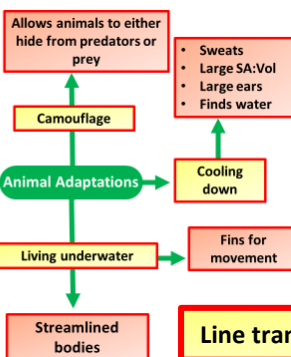
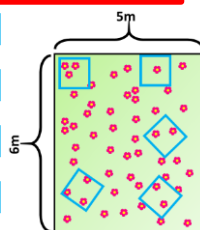
Biotic & Abiotic Factors

Biotic components are the living things that shape an ecosystem. Examples of biotic components include animals, plants, fungi, and bacteria

Abiotic factors are non-living. They include light intensity, temperature and moisture levels. The abundance and distribution of living organisms in an ecosystem are affected by abiotic factors

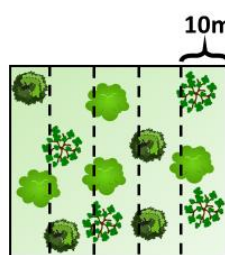
Using quadrats

A quadrat is usually a square made of wire. It may contain further wires to mark off smaller areas inside, such as 5 x 5 squares or 10 x 10 squares. The organisms underneath, usually plants, can be identified and counted. Quadrats may also be used for slow-moving animals, eg slugs



Line transects

- choose a starting point on the school field in an area where the grass is often cut
- use random numbers to generate a set of coordinates to place your first quadrat
- count the number of different plant species within this quadrat (the species richness)
- return to your starting position and repeat steps two and three a further 14 times using different random numbers
- repeat steps one to four for a part of the school field which the grass is infrequently cut
- compare your results by calculating a mean for each location

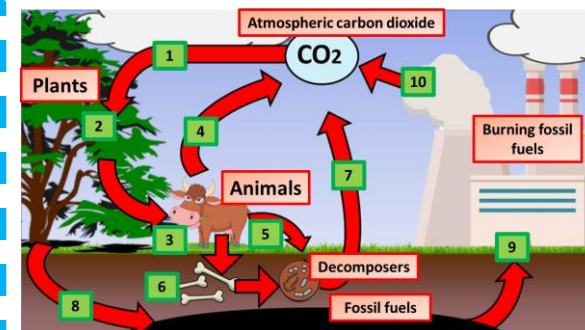


Belt transect

A transect is a line across a habitat or part of a habitat. It can be as simple as a string or rope placed in a line on the ground. The number of organisms of each species along a transect can be observed and recorded at regular intervals. The distribution of organisms in a habitat is affected by the presence of other living organisms

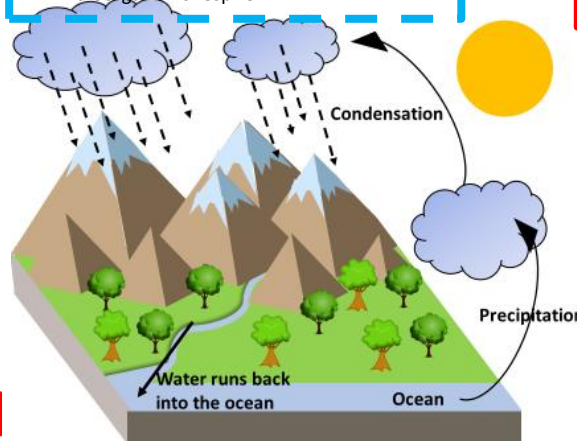
Carbon Cycle

- (green) plants photosynthesise
- photosynthesis takes in carbon dioxide
- (green) plants use carbon to make carbohydrate / protein / fat / organic compounds / named (e.g. enzymes / cellulose)
- animals eat (green) plants (and other animals)
- (green) plants respire
- animals respire
- respiration releases carbon dioxide
- (green) plants and animals die
- microorganisms decay / decompose / rot / break down / feed on dead organisms
- microorganisms respire



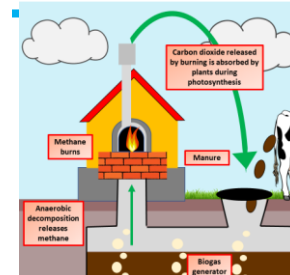
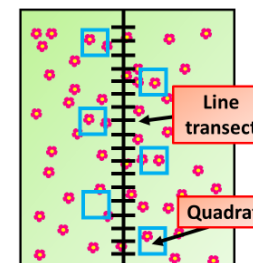
Water Cycle

The water cycle is also known as the hydrological cycle. It describes how water moves on, above, or just below the surface of our planet. Water molecules move between various locations - such as rivers, oceans and the atmosphere - by specific processes. Water can change state

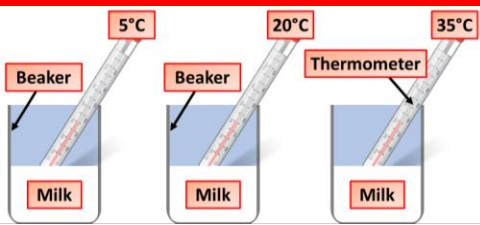


Decomposition

Decay is an essential life process, which helps to digest food and recycle materials. Bacteria and fungi are the main groups of decomposer. They release enzymes to break down compounds, so that they can absorb the nutrients. We can preserve our food, for example by freezing or canning, to remove the conditions needed for decay to happen.

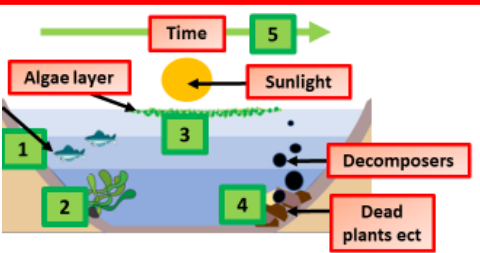


Decomposition investigation



- Place 20 cm³ of fresh milk into three beakers
- Decide the three temperatures you will investigate. Write these onto the sides of the beakers. They may be 5, 20 and 35°C.
- Use universal indicator paper or solution to determine the pH of the milk in the three beakers
- Cover each beaker in cling film and incubate at the appropriate temperature
- Use universal indicator paper or solution to determine the pH of the milk in the three beakers after 24, 48 and 72 hours

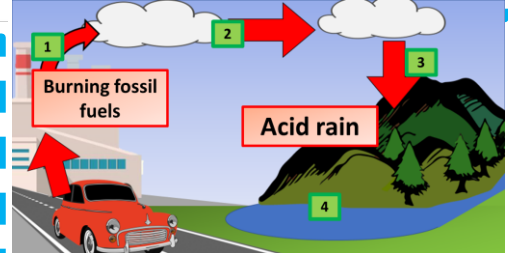
Water Pollution



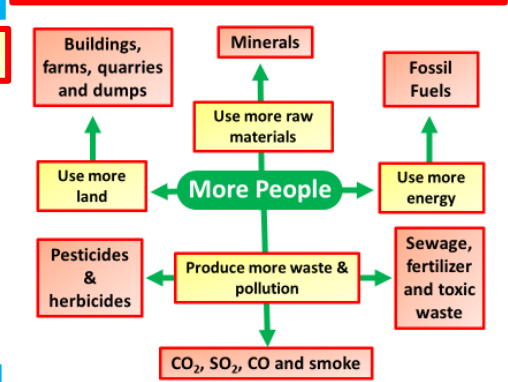
- Eutrophication - Natural eutrophication is the process by which lakes gradually age and become more productive.
- Eutrophication is a process whereby water bodies, such as lakes, estuaries, or slow-moving streams receive excess nutrients
- that stimulate excessive plant growth (algae, periphyton attached algae, and nuisance plants weeds).
- This enhanced plant growth, often called an algal bloom,
- reduces dissolved oxygen in the water when dead plant material decomposes and can cause other organisms to die.
- Nutrients can come from many sources, such as fertilizers applied to agricultural fields, golf courses, and suburban lawns; deposition of nitrogen from the atmosphere; erosion of soil containing nutrients; and sewage treatment plant discharges.

Acid Rain

Combustion of fossil fuels and other fuels releases carbon dioxide. This contributes to the greenhouse effect and leads to global warming. It also releases sulfur dioxide and nitrogen oxides which can cause acid rain. Air pollution can also be caused by tiny particulates from smoke which can cause smog.



Waste Management

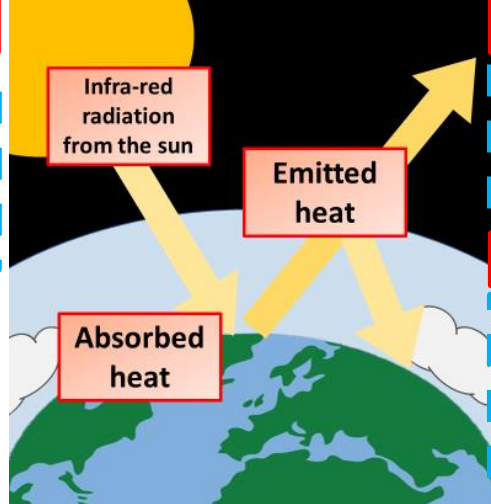


Biodiversity

'Biological diversity' means the variability among living organisms from all sources including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are a part; this includes diversity within species, between species and of ecosystems.

Deforestation

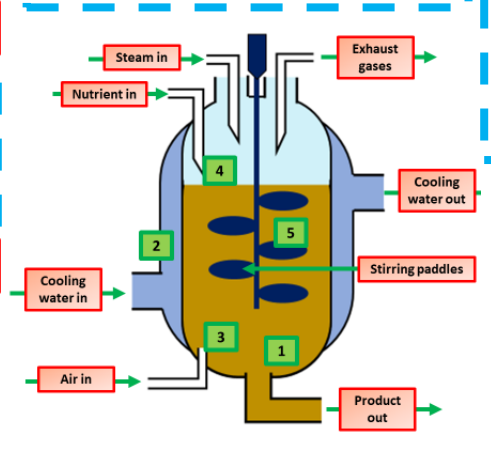
Deforestation has some important consequences: forest habitats are destroyed. soil erosion increases, causing barren land, flooding and land slides. air pollution is caused when trees are burned after felling.



Biotechnology

A good example of this is the fungus *Fusarium* which is used to produce mycoprotein. This is protein produced from a fungus. The fungus is grown in large containers called fermenters

- The pH and temperature are maintained at the optimum
- The temperature is controlled by a water jacket that surrounds the whole fermenter
- Sterile oxygen is added to make sure that aerobic respiration occurs
- A food source like glucose syrup is added
- The mixture inside is stirred to make sure all the oxygen and nutrients are equally distributed

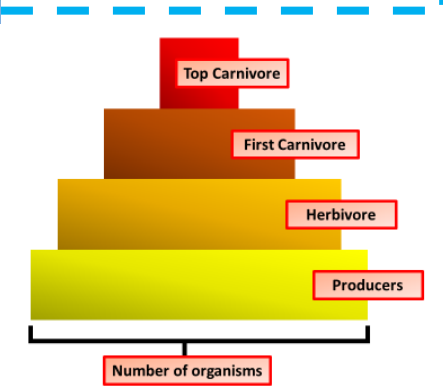


Global Warming

Global warming is projected to have a number of effects on the oceans. Ongoing effects include rising sea levels due to thermal expansion and melting of glaciers and ice sheets, and warming of the ocean surface, leading to increased temperature stratification.

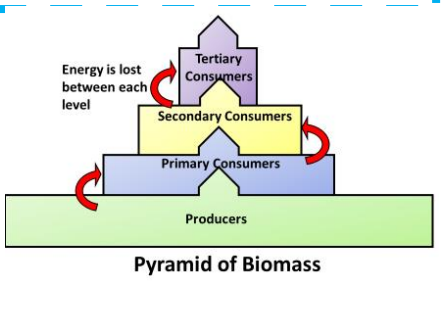
Trophic Levels

The trophic level of an organism is the position it occupies in a food chain. A food chain is a succession of organisms that eat other organisms and may, in turn, be eaten themselves. The trophic level of an organism is the number of steps it is from the start of the chain.



Pyramids of Biomass

A pyramid of biomass is a more accurate indication of how much energy is passed on at each trophic level. Biomass is the mass of living material in each organism multiplied by the total number of organisms in that trophic level



B7 Ecology AQA Trilogy

Explain what is meant by a producer in a food chain or web

Give examples of structural adaptations



RPA 7: measuring population size with quadrats
Describe how to use a quadrat to measure the distribution of daisies on the school field



Define the terms:
Habitat

Population

Community

Ecosystem

Define the terms:
Primary consumer

Secondary consumer

Tertiary consumer

Give examples of behavioural adaptations



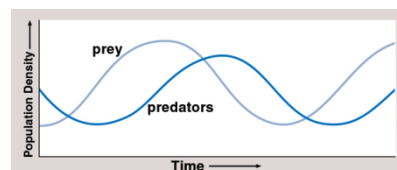
RPA 7: measuring population size with transects
Describe how to use a transect and a quadrat to measure the distribution of organisms on the seashore.



List abiotic (non-living) factors

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Describe the relationship between predators and prey



Give examples of functional adaptations



Anna counted the number of daisies in 7 quadrats. Her results were: 18, 20, 22, 23, 23, 23, 25.

Calculate the:

Mean

Mode

Median



List biotic (living) factors

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Define the terms:
Predator

Prey

Biomass

Give examples of extremophiles

Students used 0.5m² quadrats to randomly sample daisies on the school field. Their mean number of daisies per quadrat was 10.5 and the field has an area of 800m². Estimate the population of daisies on the field



B7 Ecology AQA Trilogy

Define the term biodiversity



How can pollution occur in the air



Describe some of the biological consequences of global warming

Name the two gases that contribute to global warming

How is carbon removed from the atmosphere?

List 3 ways that CO₂ is released into the atmosphere

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Explain why it is important to maintain biodiversity

How are humans reducing the land available for animals and plants?

Describe positive human interactions in an ecosystem

Define the terms:
Evaporation

Condensation

How can pollution occur in water



Why have humans completed large-scale deforestation?



Describe negative human interactions in an ecosystem

Define the terms
Transpiration

Precipitation

How can pollution occur on land

Why have humans destroyed peat and what is a problem associated with burning peat.



Give examples of programmes put in place to reduce the negative impacts of humans on ecosystems and biodiversity

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